

Modern Technology Directive: This application relies on modern web standards, JIT-accelerated backend execution, and native SQL vector indexing. We require modern software versions (PHP 8.3/8.4, MariaDB 11.8+, HTTP/3 QUIC) rather than legacy hosting defaults.

1. ARCHITECTURE & PLATFORM OVERVIEW

Application Stack

CCRM is a real-time business management system combining an ultra-fast client-side SPA (React 19, TypeScript, Vite) with an asynchronous PHP REST backend and a dual relational + vector database layer. It is built to support offline synchronization, real-time WebPush notifications, and native RAG (Retrieval-Augmented Generation) semantic vector search.

2. MODERN HARDWARE & HYPERVISOR SPECIFICATIONS

Hardware Baseline

COMPONENT	STANDARD DEPLOYMENT (UP TO 15 USERS)	HIGH-LOAD / ENTERPRISE (15–50+ USERS)
CPU Architecture	2 to 4 vCPUs (Modern AMD EPYC Genoa/Bergamo or Intel Xeon 4th/5th Gen with AVX2/AVX-512)	4 to 8 dedicated vCPUs (High single-thread frequency >3.4 GHz with AVX-512 vector support)
Memory (RAM)	4 GB DDR5 ECC RAM	8 GB – 16 GB DDR5 ECC RAM
Storage Array	25 GB+ Enterprise PCIe 4.0/5.0 NVMe SSD (High IOPS, RAID 10 or Ceph)	80 GB+ Enterprise PCIe 4.0/5.0 NVMe SSD (Dedicated high-IOPS storage)
Network & CDN	1 Gbps uplink, dedicated static IPv4 + IPv6, TLS 1.3	1 Gbps – 10 Gbps uplink, DDoS mitigation, low latency routing
Operating System	Modern Linux: Ubuntu 24.04 LTS (Noble Numbat) or Debian 12 (Bookworm) with Kernel 6.8+	Ubuntu 24.04 LTS / Container host (Docker Engine 27+ / Containerd)

3. WEB SERVER & MODERN TRANSPORT PROTOCOLS

HTTP/3 • TLS 1.3

Web Server Engine

APACHE 2.4.60+ / LITESPEED / NGINX

- HTTP/2 and HTTP/3 (QUIC):** Required for instantaneous PWA asset multiplexing.
- Brotili & Gzip Compression:** `mod_brotli` active for static JavaScript/CSS.
- Modules:** `mod_rewrite` , `mod_headers` , `mod_authz_core` .
- AllowOverride All:** Strictly enabled on docroot for `.htaccess` router guards.
- `DirectoryIndex index.html index.php`

SSL / TLS & Security Headers

STRICT MANDATORY

Automated **SSL / TLS 1.3** certificate (Let's Encrypt / ZeroSSL with modern ECDSA P-256 keys). Mandatory for Progressive Web App (PWA) installation, background Service Worker sync, and RFC 8291 Web Push APIs.

4. NEXT-GEN PHP RUNTIME (PHP 8.3 / 8.4 WITH JIT)

PHP 8.3+ Preferred

Target version: **PHP 8.3 or PHP 8.4** (Minimum supported: PHP 8.2). Required extensions & settings:

PHP EXTENSION	CATEGORY	PURPOSE IN CCRM STACK
<code>pdo_mysql</code>	Database	PDO connector for MySQL 8 & MariaDB 11.8 with prepared statements & DDL migrations.
<code>imap</code>	Mail Client	Direct IMAP synchronization, TLS mail socket negotiation, and multi-part attachment parsing.
<code>zip</code> (ZipArchive)	Compression	Asynchronous backup creation, bulk data exports, and packaging client invoices.
<code>curl</code>	Network API	High-performance cURL client for OpenAI/LLM streaming, SuperFaktura, iDoklad, and RegisterUZ.
<code>openssl</code>	Crypto / WebPush	RFC 8291/8292 VAPID ECDSA P-256 signing, AES-256-GCM encryption, and JWT validation.
<code>mbstring</code>	Multilingual	UTF-8 text processing across multi-lingual datasets (Slovak, Hungarian, Czech, English).
<code>fileinfo</code>	File Safety	Server-side MIME inspection to prevent malicious executable file uploads.
<code>opcache</code>	JIT Acceleration	OPcache with JIT compiler enabled (<code>opcache.jit=tracing</code> , <code>opcache.jit_buffer_size=128M</code>).

Target PHP Configuration Parameters (php.ini or .user.ini)

`memory_limit = 512M``upload_max_filesize = 128M``post_max_size = 140M`

`max_execution_time = 180``max_file_uploads = 50``opcache.enable = 1`

PARAMETER	TECHNICAL SPECIFICATION
Engine & Version	MySQL 8.4 LTS / MySQL 8.0+ OR MariaDB 11.8+ (Latest stable release)
Character Encoding	Full UTF-8 4-byte charset: utf8mb4 with collation utf8mb4_unicode_ci (or utf8mb4_0900_ai_ci on MySQL 8)
Database Permissions	SELECT, INSERT, UPDATE, DELETE, CREATE, DROP, ALTER, INDEX, REFERENCES (needed for automated DDL migration engine)

Mandatory MariaDB Version for Native Vector RAG: CCRM's internal semantic retrieval engine utilizes native SQL vector capabilities introduced in **MariaDB 11.8.0 or newer**. Older MariaDB series (10.5, 10.6, 10.11 LTS, 11.4 LTS) and standard MySQL do NOT have native vector datatypes or cosine similarity functions.

FEATURE / REQUIREMENT	TECHNICAL REQUIREMENT & VERIFICATION DETAILS
Exact Database Engine	MariaDB 11.8.0 or higher (Rolling series with native vector storage plugin enabled)
Native Vector Datatype	Native VECTOR(<dim>) column type (e.g. VECTOR(1536) for OpenAI text-embedding-3 or VECTOR(384) for local MiniLM)
Vector Search Functions	Hardware-accelerated VEC_DISTANCE_COSINE(), VEC_DISTANCE_EUCLIDEAN(), VEC_DISTANCE()
Index Architecture	Native HNSW (Hierarchical Navigable Small World) index support for sub-millisecond approximate nearest neighbor search
CPU Acceleration	Host CPU must expose AVX2 or AVX-512 vector instructions to the database process for high-speed dot-product calculations
Deployment Model	Option A (Unified MariaDB 11.8+): Single instance powering both relational CRM records & native vector tables. Option B (Dedicated Vector Sidecar): MariaDB 11.8+ running on dedicated port (e.g. 3309) alongside MySQL. Option C (Alternative Fallback): Dedicated Qdrant sidecar container (v1.9+, port 6333) or external Pinecone Cloud.

Host Firewall Verification: Outbound socket connections must not be filtered. The following outbound ports are required:

TARGET PORT	PROTOCOL	SERVICE & API DESCRIPTION
Port 443 (Outbound)	HTTPS / TLS	LLM APIs (OpenAI, Anthropic, OpenRouter), SuperFaktura, iDoklad, RegisterUZ, ARES, Pinecone.
Port 993 (Outbound)	IMAP over SSL	Real-time synchronization of client emails into user CRM mailboxes.
Port 465 / 587 (Outbound)	SMTP / TLS	Outbound transactional notifications, task reminders, and automated workflow dispatches.

Automated deployment via php ccrm update over SSH:

- OpenSSH:** Ed25519 key authentication.
- Git 2.40+:** Fast-forward release synchronization.
- Composer 2.7+:** PHP dependency autoloader optimization.

Triggers asynchronous queues & reminder dispatchers every 1 min:

```
* * * * * curl -s -f "https://crm.yourdomain.sk/api/cron.php?token=TOKEN" >/dev/null 2>&1
```

☐ Modern PHP runtime: **PHP 8.3+ or 8.4** with OPcache JIT enabled

☐ Mandatory PHP modules active: pdo_mysql , imap , zip , curl , openssl , mbstring , fileinfo

☐ Modern Web Server with **HTTP/2 & HTTP/3 (QUIC)** , Brotli compression, and AllowOverride All

☐ **RAG Vector Database:** Native **MariaDB 11.8+** (with VECTOR datatype & HNSW) OR Qdrant container support

☐ Host CPU vector extensions enabled (AVX2 / AVX-512) for vector distance calculations

☐ High-speed PCIe NVMe SSD storage with automated daily snapshot backups

☐ Unrestricted outbound connectivity on Port 993 (IMAP) and Port 443 (HTTPS)

☐ Modern Linux OS (Ubuntu 24.04 LTS / Debian 12) with SSH shell and Git 2.40+